A Network No-Arbitrage Theory of Bid–Ask Spreads, Arbitrage Premiums, and Trade Compression in the International Currency–Trading Architecture

Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

This paper develops a graph-theoretic no-arbitrage theory for the international currency-trading architecture.

The framework begins with:

4 nations,

5 reserve currencies,

20 nation–currency pairs,

190 trade relationships,

and

380 currency traders.

A complete interaction graph K_{20} is introduced to represent all potential monetary interactions.

Bid and ask prices are defined on graph edges and a global no-arbitrage condition is obtained through conservative log-price fields.

An arbitrage-premium mechanism is then proposed which imposes positive trading costs on trades participating in positive-arbitrage cycles.

The resulting framework naturally compresses the global trading network while preserving monetary connectivity.

The theory unifies international finance, market microstructure, graph theory, economics, computer science, data science, artificial intelligence, political economy, and geopolitical strategy.

The paper ends with "The End"

Contents

1 Introduction

Classical no-arbitrage theory studies consistency among prices.

The present work studies consistency among prices, networks, and traders simultaneously.

The central object is the international currency-trading graph

$$K_{20}.$$

The objective is to construct a system in which:

1. arbitrage opportunities are measurable,
2. arbitrage costs are endogenous,
3. market complexity is controllable,
4. network compression emerges naturally.

2 The International Trading Network

Definition 1. *Let*

$$V = P_{ij},$$

where

$$P_{ij} = (N_i, C_j).$$

The graph contains

$$|V| = 20$$

vertices.
Every pair of vertices is potentially tradable.
Therefore

$$G = K_{20}.$$

Proposition 1. *The network contains*

$$190$$

trade links.

Proof. A complete graph on twenty vertices possesses

$$\binom{20}{2} = 190$$

edges.

□

3 Bid and Ask Operators

For edge

$$e = (u, v),$$

define:

$$B_{uv}$$

as the bid quote and

$$A_{uv}$$

as the ask quote.
Necessarily

$$A_{uv} \geq B_{uv}.$$

Definition 2. *The bid–ask spread is*

$$s_{uv} = A_{uv} - B_{uv}.$$

Definition 3. *The midprice is*

$$M_{uv} = \frac{A_{uv} + B_{uv}}{2}.$$

Thus

$$A_{uv} = M_{uv} + \frac{s_{uv}}{2},$$

and

$$B_{uv} = M_{uv} - \frac{s_{uv}}{2}.$$

4 Logarithmic Price Geometry

Define

$$r_{uv} = \log M_{uv}.$$

Definition 4. *The log-price field is the collection*

$$r_{uv}.$$

5 Cycle Arbitrage

Consider a cycle

$$C = u \to v \to w \to u.$$

Definition 5. *Cycle arbitrage profit is*

$$\Pi(C) = r_{uv} + r_{vw} + r_{wu}.$$

Definition 6. *Positive arbitrage is*

$$\Pi^+(C) = \max(\Pi(C), 0).$$

6 Global No-Arbitrage

Theorem 1 (Network No-Arbitrage)**.** *The trading network is arbitrage-free if and only if*

$$\sum_{e \in C} r_e = 0$$

for every cycle C.

Proof. If every cycle sum vanishes, the log-price field is conservative.
Therefore there exists a potential function

$$\phi : V \to \mathbb{R}$$

such that

$$r_{uv} = \phi(v) - \phi(u).$$

Hence no cycle can generate positive profit.
Conversely, any nonzero cycle sum produces an arbitrage opportunity. □

Corollary 1. *There exists a global monetary potential*

$$\phi.$$

7 Global Numeraire Potential

Choose a numeraire node

$$P_{11}.$$

Set

$$\phi(P_{11}) = 0.$$

Then

$$M_{uv} = \exp\left(\phi(v) - \phi(u)\right).$$

All exchange rates become generated by a single global monetary surface.

8 Spread Formation

8.1 Inventory Risk

Let

$$I_e$$

denote inventory risk.

8.2 Latency Risk

Let

$$L_e$$

denote execution latency.

8.3 Capital Cost

Let

$$C_e$$

denote funding cost.

8.4 Adverse Selection

Let

$$A_e$$

denote information asymmetry.

Theorem 2 (Spread Decomposition). *The spread satisfies*

$$s_e = \alpha I_e + \beta L_e + \gamma C_e + \delta A_e.$$

9 Arbitrage Premium

Definition 7. *For every edge e define*

$$\tau_e = \kappa \sum_{C \ni e} \Pi^+(C),$$

where

$$\kappa > 0.$$

Remark 1. *The premium is zero when no positive arbitrage exists.*

10 Effective Spread

Definition 8. *The effective spread becomes*

$$s_e^* = s_e + \tau_e.$$

Hence

$$A_e^* = A_e + \frac{\tau_e}{2},$$

and

$$B_e^* = B_e - \frac{\tau_e}{2}.$$

11 Arbitrage Suppression

Theorem 3 (Arbitrage Suppression)**.** *The premium increases transaction costs on arbitrage-producing edges.*

Proof. For positive arbitrage,

$$\Pi^+(C) > 0.$$

Hence

$$\tau_e > 0.$$

Therefore

$$s_e^* > s_e.$$

Transaction costs increase.

□

12 Trade Volume Dynamics

Suppose

$$V_e = V_0 e^{-\lambda s_e^*}.$$

Substituting,

$$V_e = V_0 e^{-\lambda(s_e + \tau_e)}.$$

Proposition 2. *Trade volume decreases as arbitrage premiums increase.*

Proof. Differentiating gives

$$\frac{\partial V_e}{\partial \tau_e} = -\lambda V_e < 0.$$

□

13 Network Compression

Initially

$$|E| = 190.$$

The graph is fully connected.

Definition 9. *Let*

$$E_t$$

denote active trading edges.

Then

$$E_t \leq 190.$$

Increasing arbitrage penalties removes economically unnecessary edges.

14 Minimal Connectivity

A connected graph on twenty vertices requires

$$20 - 1 = 19$$

edges.

Theorem 4 (Compression Bound)**.** *The active trading network satisfies*

$$19 \leq |E_t| \leq 190.$$

Proof. The lower bound is the minimum edge count for connectivity.
The upper bound is the complete graph. □

15 Spectral Spread Theory

Let

$$L = D - A$$

denote the graph Laplacian.
Let

$$s = (s_1, \ldots, s_{190})^T.$$

Definition 10. *Market stress is*

$$\mathcal{E} = s^T L s.$$

Proposition 3. *Higher values of*

$$\mathcal{E}$$

correspond to greater market fragmentation.

16 The Unified Spread Equation

Combining all effects yields

$$s_e = \alpha I_e + \beta L_e + \gamma C_e + \delta A_e + \eta |\nabla_G \phi|_e + \zeta \lambda_2(G).$$

Term	Interpretation
I_e	Inventory Risk
L_e	Latency Risk
C_e	Capital Cost
A_e	Adverse Selection
$\lvert\nabla_G \phi\rvert$	Monetary Gradient
$\lambda_2(G)$	Connectivity Effect

Table 1: Spread Components

17 Economics

The arbitrage premium acts as a Pigouvian correction.

Arbitrage-producing trades impose instability costs on the network.

The premium internalizes these costs.

18 Finance

The framework extends traditional bid–ask spread theory by introducing:

1. graph topology,
2. cycle arbitrage,
3. endogenous penalties,
4. network compression.

19 Political Science and International Relations

Monetary influence is transmitted through the trade network.

Reducing arbitrage-intensive links may alter:

1. reserve flows,
2. financial alliances,
3. monetary blocs,
4. geopolitical influence.

20 Warfare Economics

Reserve currencies finance strategic capabilities.

Reducing arbitrage leakage increases the efficiency of monetary power.

The premium therefore acts as a monetary-defense mechanism.

21 Computer Science

The premium functions as a regularization operator.

The optimization problem becomes

$$\min \Big(\text{Transaction Cost} + \kappa \mathcal{A} \Big),$$

where

$$\mathcal{A} = \sum_C \Pi^+(C).$$

22 Artificial Intelligence

The framework naturally supports:

1. graph neural networks,
2. reinforcement learning,
3. multi-agent systems,
4. market simulations.

The arbitrage premium acts as a reward penalty within the learning process.

23 Philosophy

The theory suggests that money is simultaneously:

1. a price system,
2. an information system,
3. a network,
4. a geometric object.

Arbitrage corresponds to curvature.
No-arbitrage corresponds to flatness.
The premium acts as a force driving the monetary system toward geometric consistency.

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Glossary

K_{20} Complete interaction graph.

Bid Price Highest purchase price.

Ask Price Lowest selling price.

Spread Difference between ask and bid.

Arbitrage Riskless profit opportunity.

Arbitrage Premium Penalty on arbitrage-producing trades.

Compression Reduction in active trade links.

Monetary Potential Global price-generating function.

Laplacian Graph operator measuring connectivity.

Spectral Stress Spread-based fragmentation measure.

The End